



From Logs to Insights: Your First Steps in Splunk

By
Sathyaprabha Gunasekaran

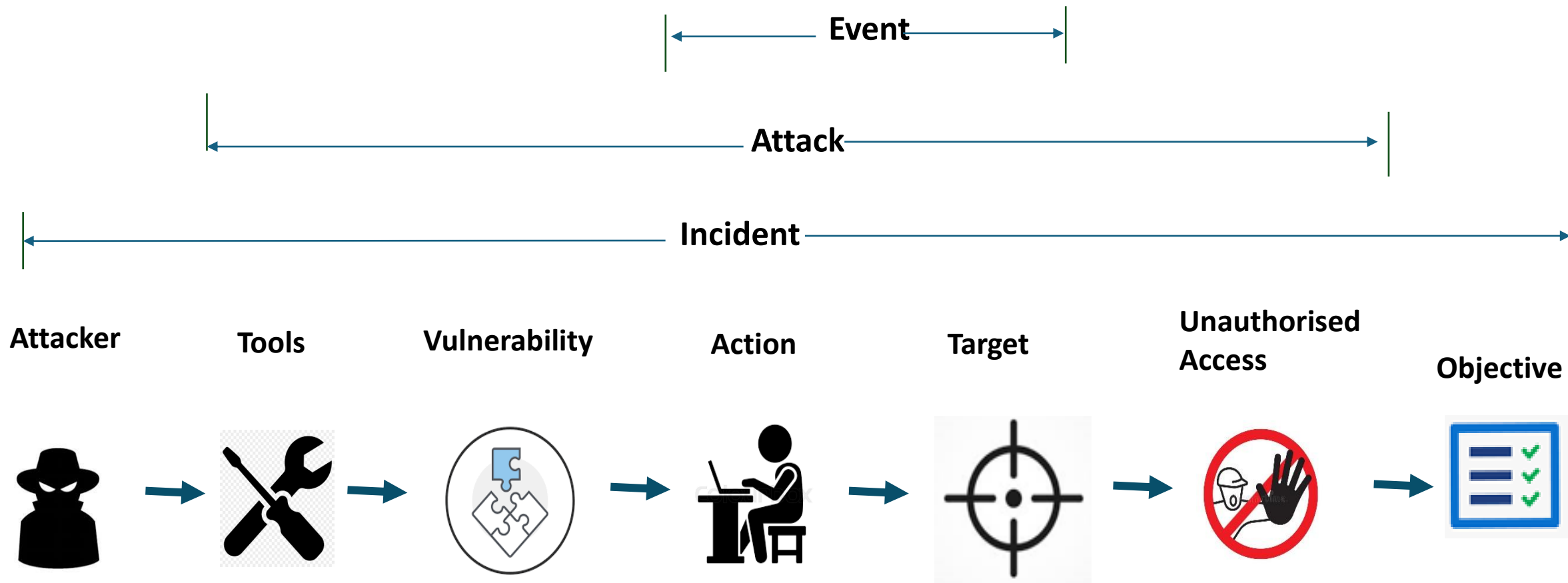
Date – Nov 14, 2025

Agenda

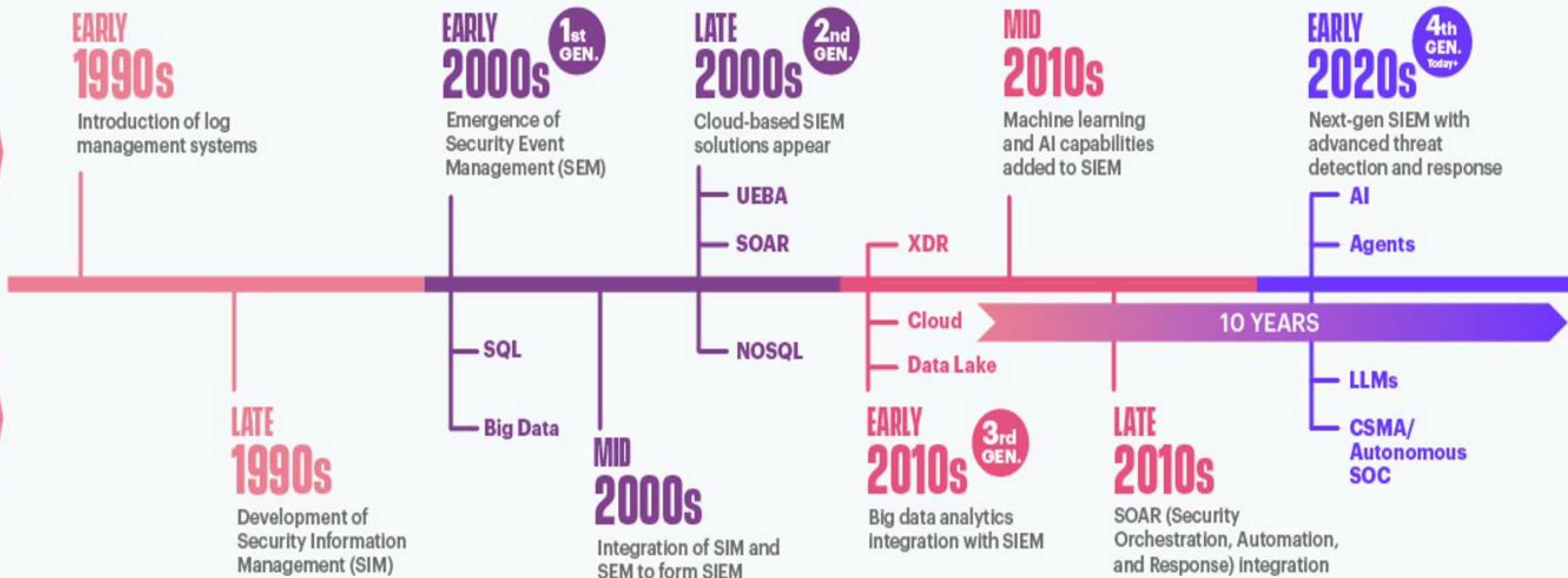
- ❖ Cyber Defense Terms
- ❖ Security Incident & Event Management (SIEM)
- ❖ SIEM Architecture
- ❖ Splunk Architecture (Single & Multi Tier)
- ❖ Splunk Components
- ❖ Log Source Integration Methods
- ❖ SIEM Workflow with Examples
- ❖ Threat Intelligence



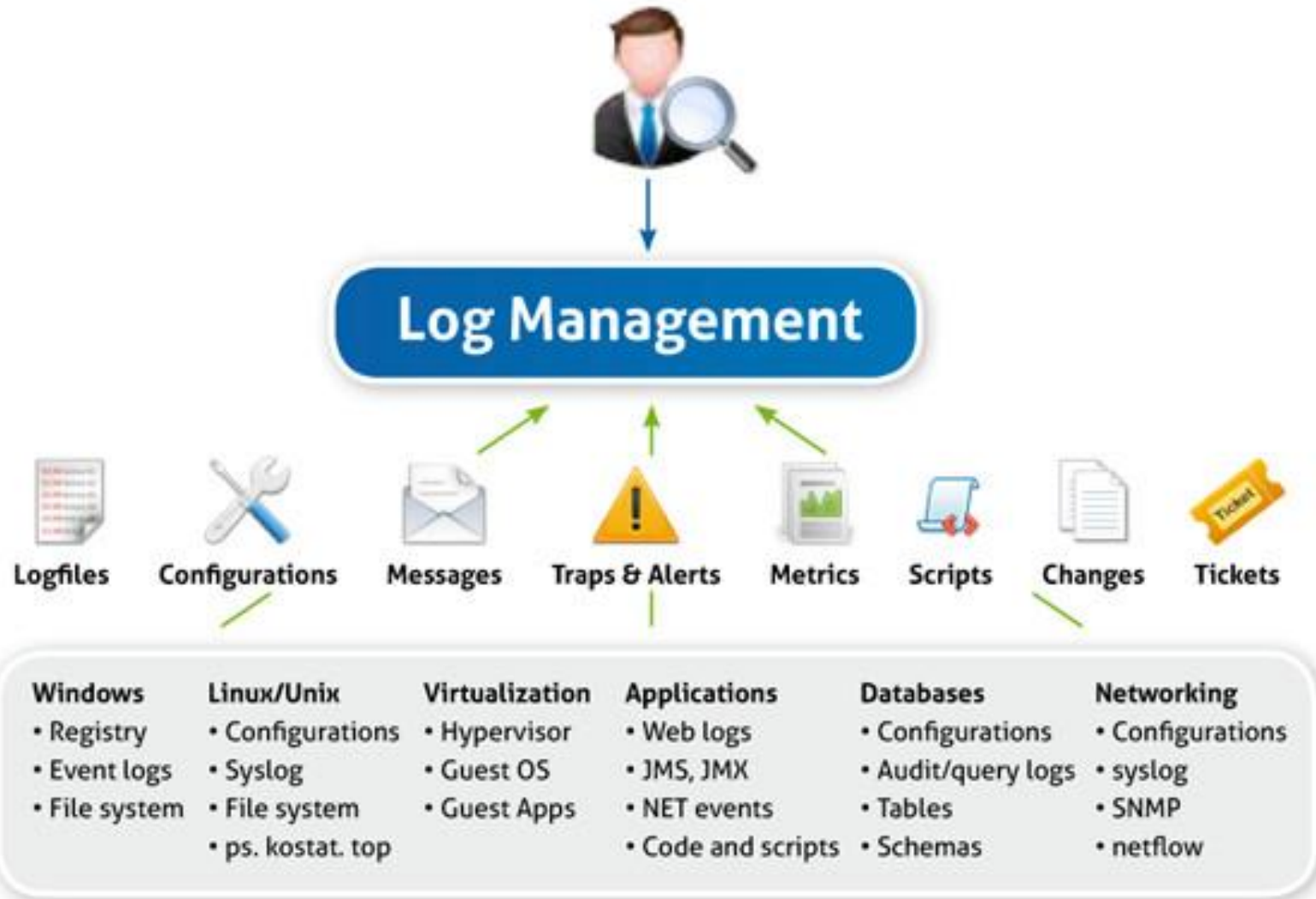
Cyber Security Terms



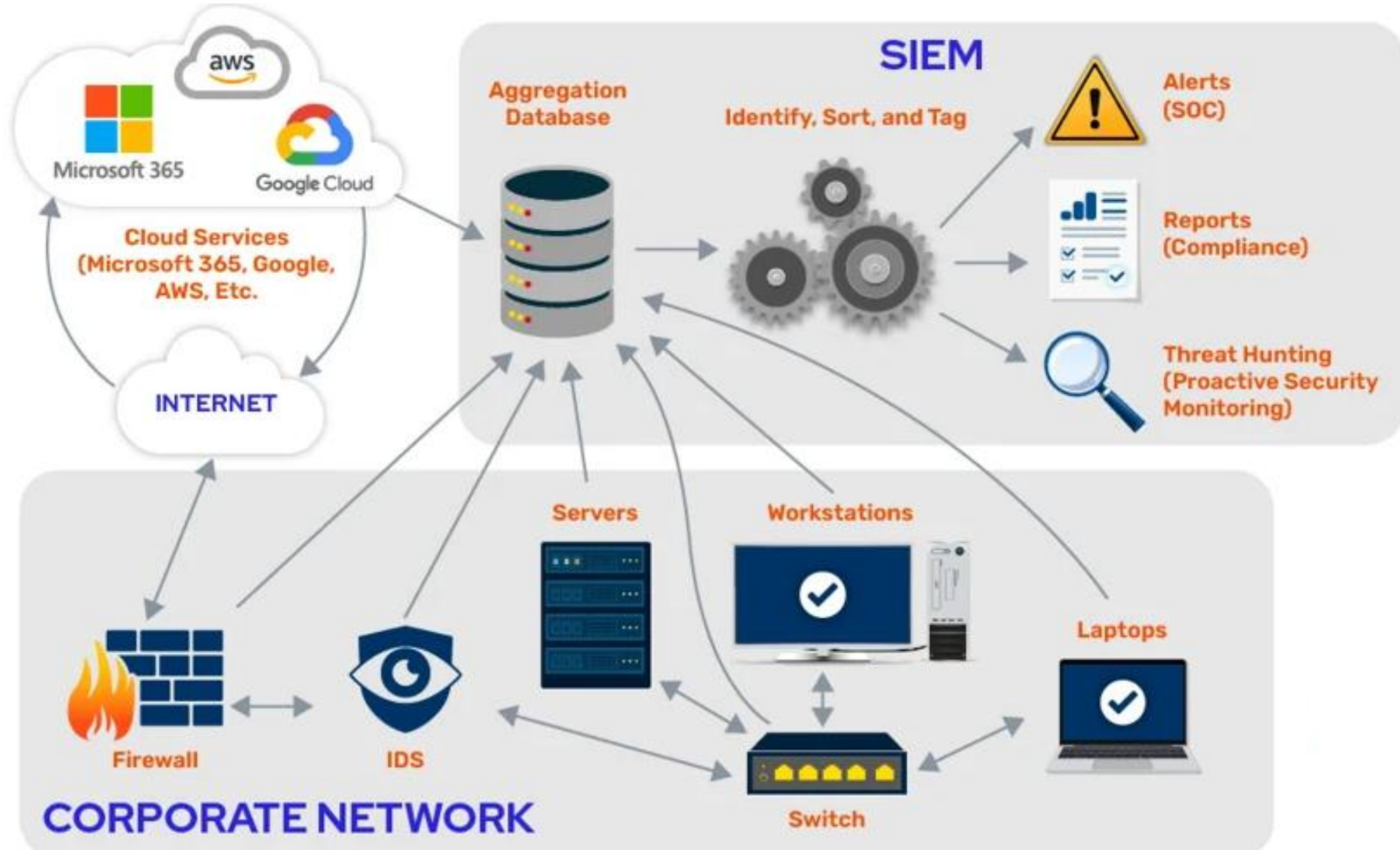
Evolution of SIEM



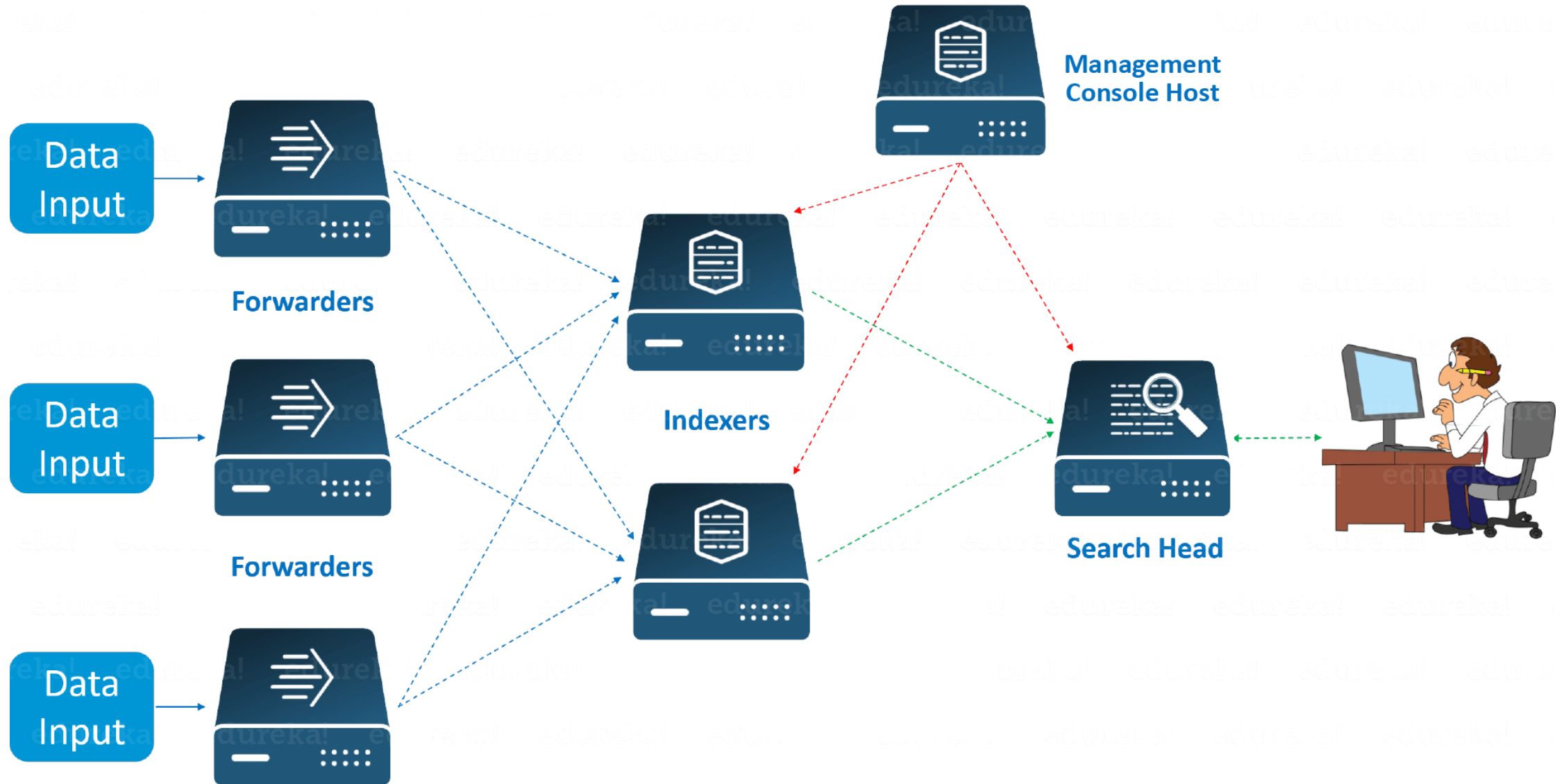
SIEM – Security Incident and Event Management



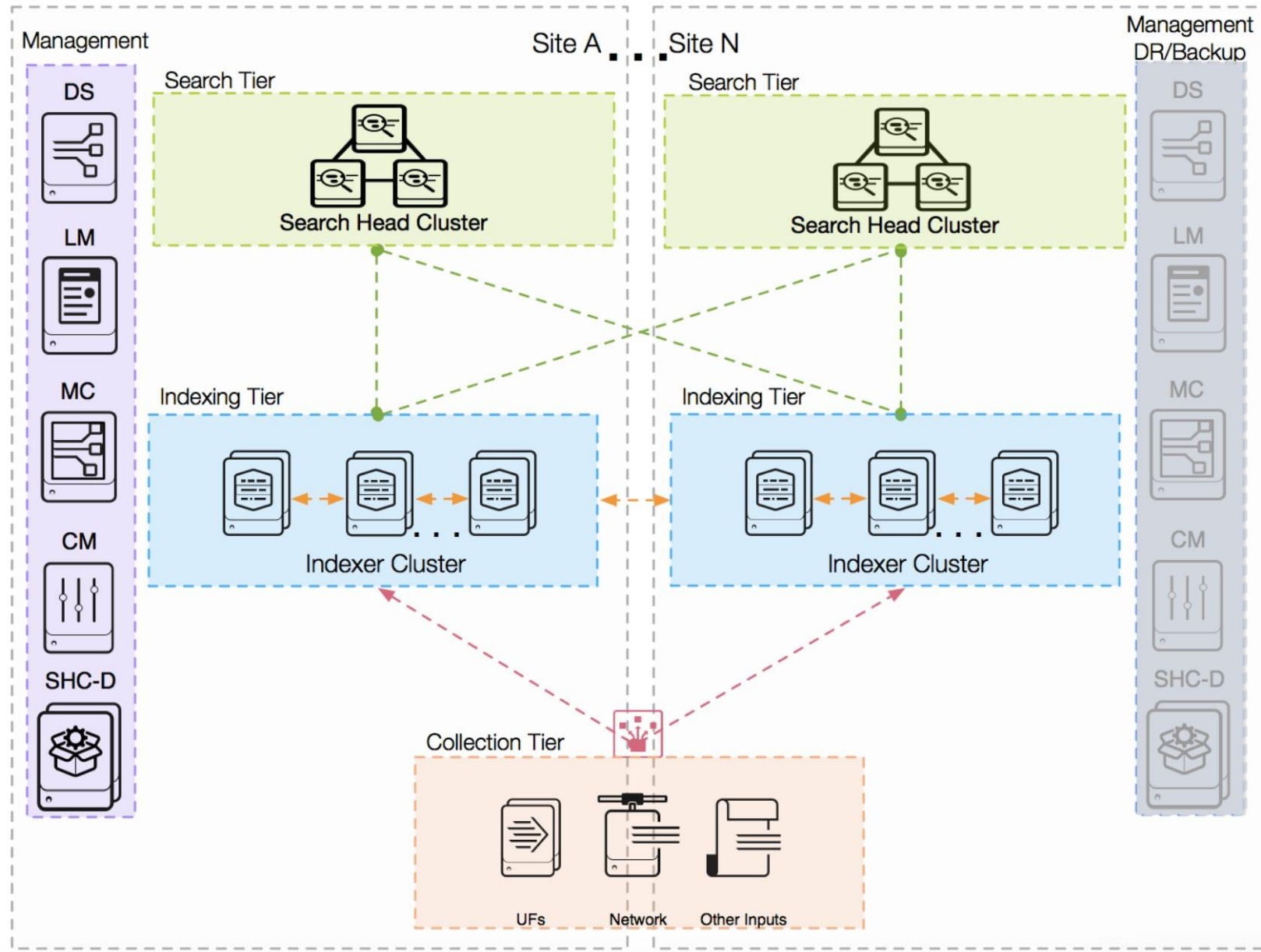
SIEM – Security Incident and Event Management



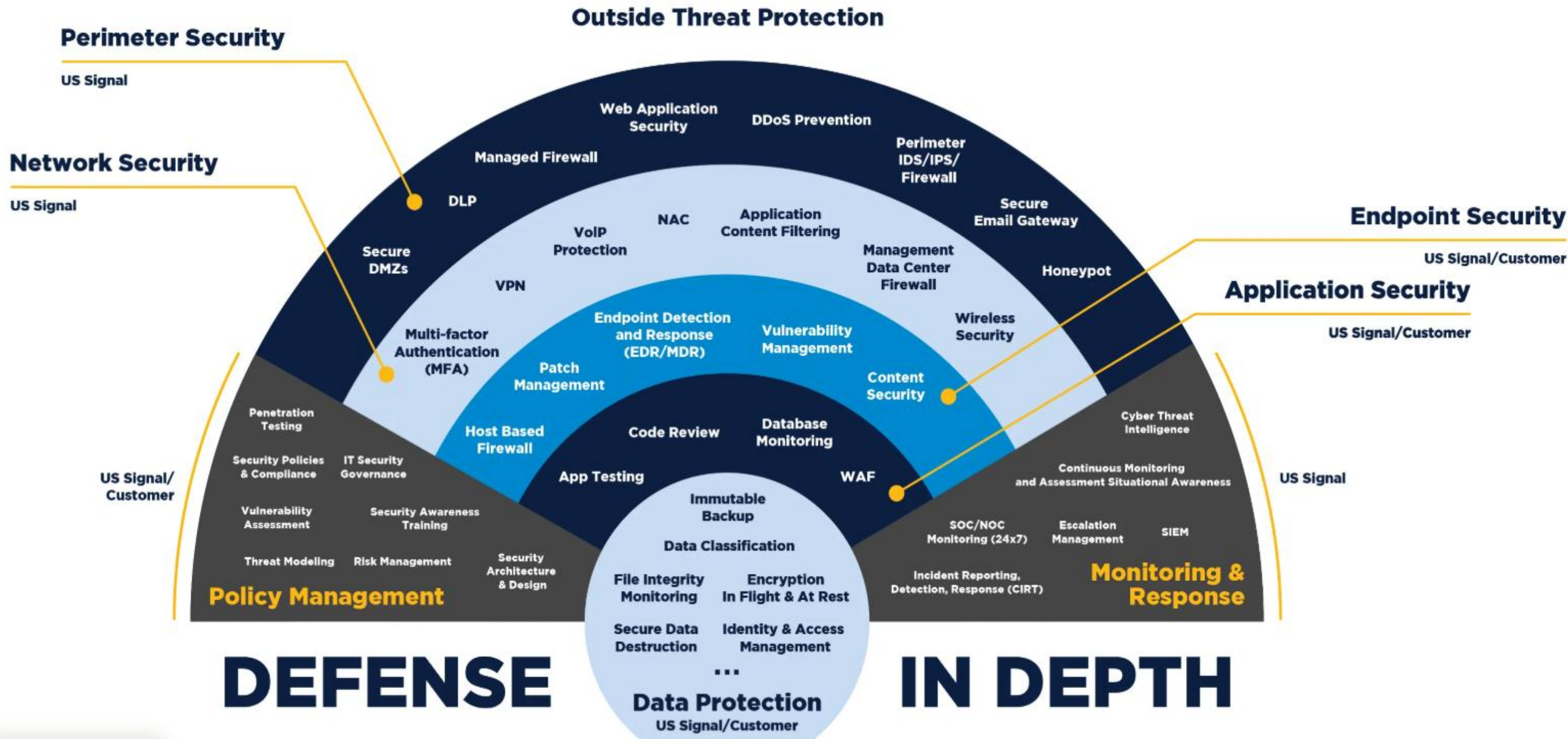
Splunk Architecture



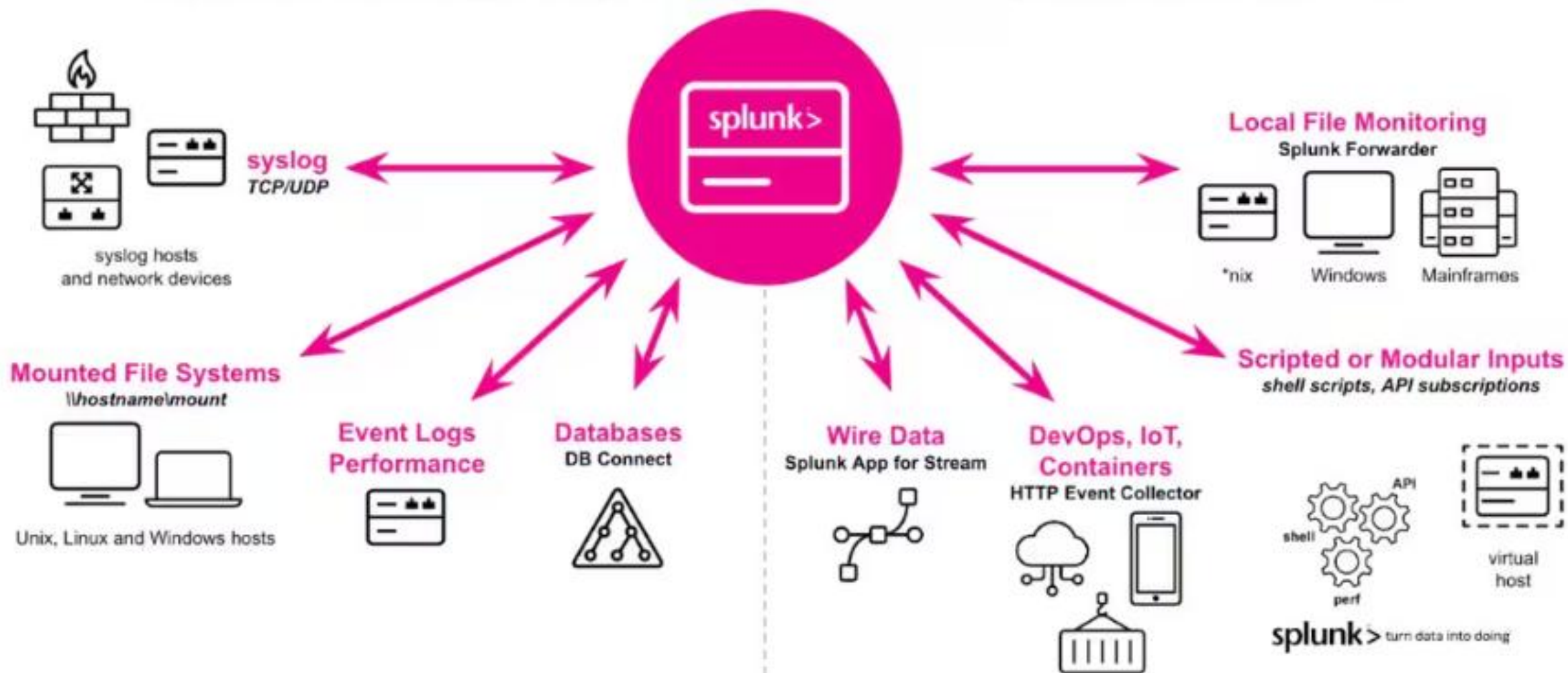
Splunk Architecture



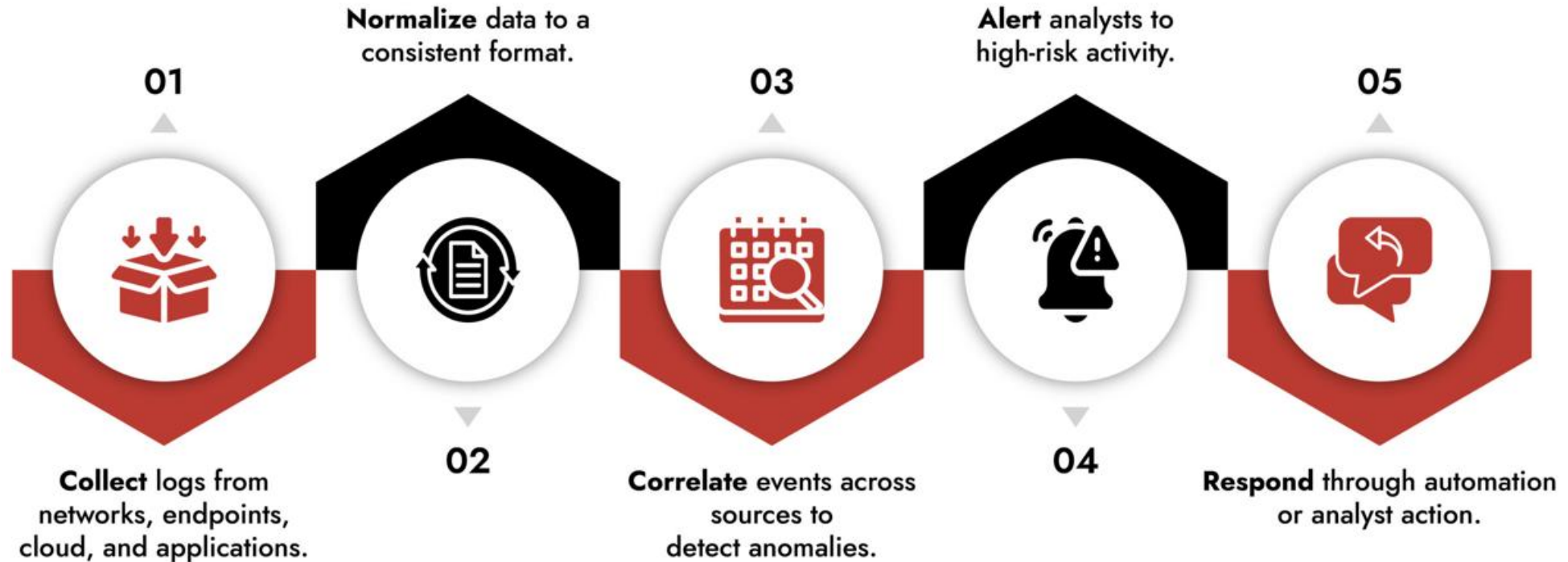
Log Sources



Log Source Integration Methods



SIEM Workflow



Integration Methods:

- Syslog – Port 514
- API through Keys
- Direct Connectors

Normalizing the Data:

- Log Parsing
- Normalization
- Event Type Mapping

Correlating the events:

- Single Event + Other Events
- Correlation
- Alert Query Mapping

SOC – Alert Triage

- Severity & Priority
- Service Level Agreement
- Investigation
- Categorization



SIEM – Security Incident and Event Management

Raw Log

<34> 2025-11-13 12:00 host1 ssh 1234 - -
Failed password for invalid user admin
from 192.168.1.10 port 22 ssh2

Parsed Log

- **Source IP:** 192.168.1.10
- **User:** admin
- **Event type:** SSH Login Failure

Normalized Field

- Src_ip
 - Client_ip
 - From_address
- ➡ source_ip
-
- UserID
 - Usr_id
 - uid
 - User_account
- ➡ User_id

Events to Correlate

Event 1 :- A user “Sathya” login failure at 1 AM
at host1 from 192.168.1.10

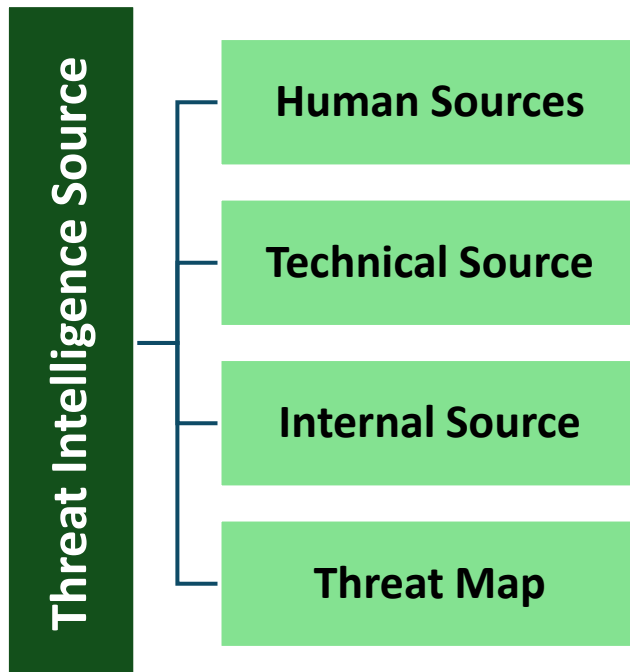
Event 2:- A user “Sathya” login failure at 1.05
AM at host2 from 192.168.1.10

Alert Query

From Authentication Logs
event_type = 'failed_login' AND INTERVAL = '10
minutes' BY username, source_ip HAVING
failed_attempts > 5

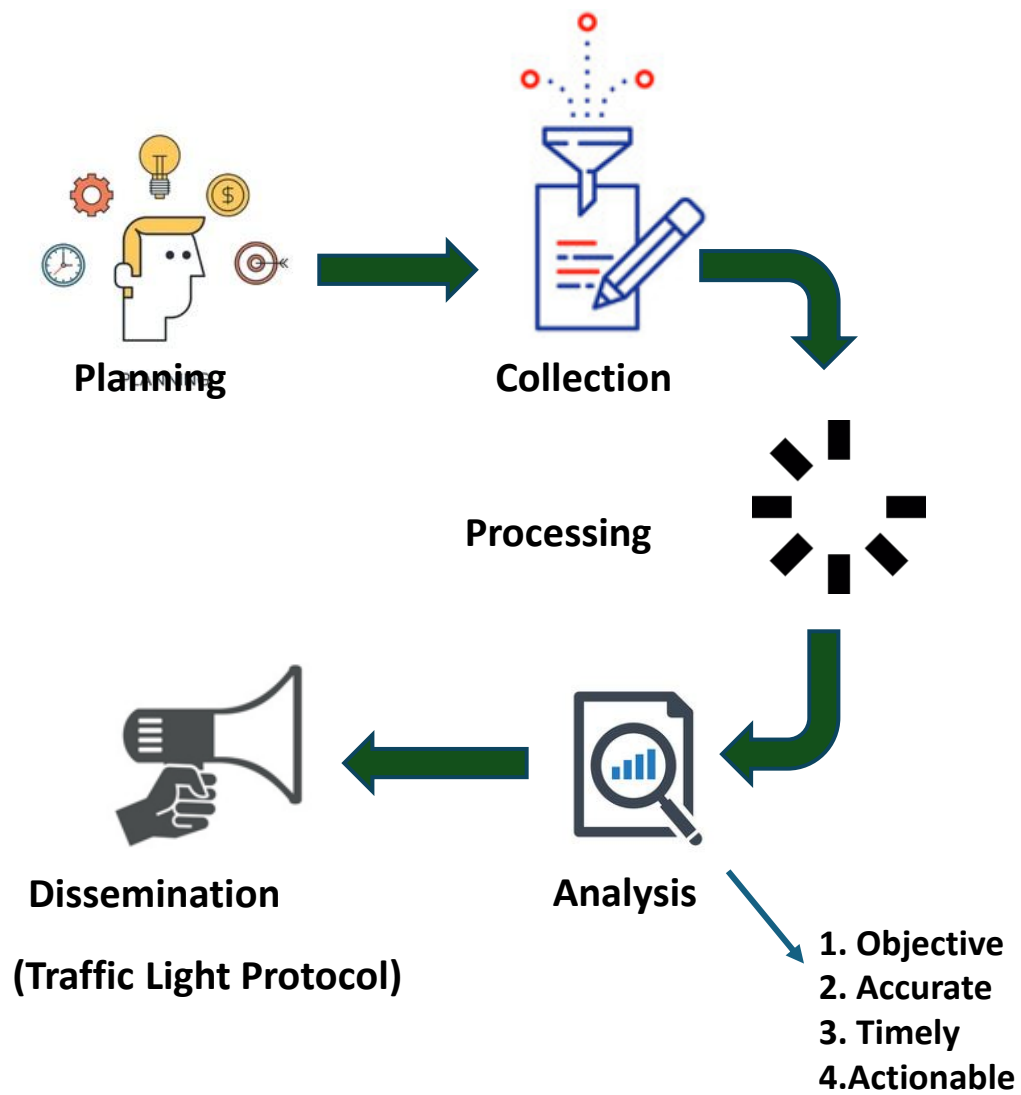
Threat Intelligence

- **Threat Intelligence** - Collection of data to understand a threat actor's motives, targets, and attack methods.
- **IOCs – Indicators of Compromise** (Hashes, Ips and Domains)



Open Source Threat
Intelligence Platforms -
OSINT

Threat Intelligence Life Cycle



Types of Threat Intelligence

- **Strategical Intelligence**
(Non-technical, Higher Mgnt)
- **Tactical Intelligence**
(Administrators)
- **Operational Intelligence**
(N/w Admin, Security Manager)
- **Technical Intelligence**
(SOC analyst, IR team)

Study Materials

Happy Learning ! 😊

Splunk

👤 **Course Catalog** – https://www.splunk.com/en_us/training/course-catalog.html?sort=Newest&filters=filterGroup1FreeCourses&locale=en_us

👤 **Basic Searching** : https://www.youtube.com/watch?v=xtyH_6iMxwA

👤 **Practical Splunk – Zero to Hero** : <https://www.youtube.com/watch?v=tMacaM1I1CI>

👤 **Splunk Use Cases** – <https://0xcybery.github.io/blog/Splunk+Use+Cases>

A top-down view of a spiral-bound notebook with a white cover, lying on a light-colored wooden surface. The notebook is open, and the words "THANK YOU!" are written in a large, bold, sans-serif font. The word "THANK" is in black, and "YOU!" is in black with a red exclamation mark. To the left of the notebook is a small, round, light-colored pot containing a green succulent plant with thick, pointed leaves. Above the notebook, a pair of gold-rimmed glasses with black temples is resting. To the right of the notebook, a silver and black ballpoint pen lies diagonally. The background is a light-colored wooden surface with visible grain.

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